

MAHEEP MOULI SHASHI

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SOCIAL LINKS

Portfolio: <https://maheep-mouli-portfolio.work/>
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SUMMARY

Architect & Computational Designer with 3+ years of experience specialising in **AI-driven design, BIM workflows, and computational methods** for architecture and urban planning. Skilled in integrating **parametric modelling, data analytics, and automation** to deliver innovative, sustainable, and efficient solutions. Adept at working across multidisciplinary teams to optimise design, performance, and execution in complex projects.

CORE COMPETENCIES

- AI & Computational Design:** Generative Design, Graph Machine Learning, LLM Integration, Automation Workflows
- BIM & Digital Twins:** Revit, Dynamo, IFC Standards, BIM 360, Clash Detection, 4D/5D BIM
- Parametric Design:** Rhino, Grasshopper, Python & C# Scripting, Environmental Analysis
- Visualisation & Presentation:** Enscape, Lumion, V-Ray, Unreal Engine, Adobe CC
- Data Analysis & Visualisation:** Python (Pandas, NumPy, Matplotlib), Power BI, Kepler.gl, GIS Mapping
- Project Management Tools:** Trello, Asana, Miro, MS Project

SKILLS

Rhino	Grasshopper
████████████████	████████████████
Revit	AutoCAD
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SketchUp	Adobe Suite
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Python	PyTorch
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FastAPI	Streamlit
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Graph ML	D3.js
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SQL	QGIS
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ArcGIS	
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EXPERIENCE

SEPTEMBER 2025-CURRENT

Volunteer Assistant | B.A.A.R. Studio – Building with Art, Architecture | Vienna

- Supporting **project development, editorial work, and grant writing** with a focus on funding applications.

- Assisting in the preparation and realization of **participatory initiatives** at the intersection of art, architecture, and social practice.
- Contributing to **community workshops and collaborations** with artisans, NGOs, and cultural practitioners.
- Involved in **Creatives for Vienna – Making Spaces**, a competition reimagining Vienna's open spaces through design and participatory programming.

MARCH 2023-MAY 2025

Architectural Designer - Remote | Atelier 24 | Remote (New Delhi, India)

- Led Rhino-Grasshopper scripting for form generation, reducing modelling time by **40%**. **Contributed to the design/documentation of 70,000+ sq. ft.** across hospitals and hotels.
- Produced **200+ technical sheets (floor plans, sections, service layouts)** using **Revit & AutoCAD**.
- Coordinated weekly meetings with MEP teams, boosting issue resolution by 30%. Managed version control systems and drawing repositories across 2 time zones.
- Transformed **client visions into stunning architectural designs**.
- Increased **project efficiency by 30%** through streamlined processes.

SEPTEMBER 2024-JUNE 2025

Thesis Researcher – Master's Project | Institute for Advanced Architecture of Catalonia (IAAC) | Barcelona

- Built an AI-based congestion index with **82% accuracy in traffic prediction**.
- Visualised city-wide urban congestion using Streamlit & Kepler.gl dashboards. Analysed **5M+ data points from 8 global cities, integrating weather, AQI, transit, and population**.
- Compared **AI-managed vs. conventional traffic systems** to model future scenarios. Paper in progress for peer-reviewed publication in urban mobility.
- Led projects from concept to completion, ensuring timely delivery.
- Utilised cutting-edge software for **precise design and visualisation**.

SEPTEMBER 2023-JUNE 2024

R&D Specialist – Circular Wood Project | WOOD-ID, IAAC | Barcelona

- Developed **ML tools for circular wood detection, nesting, and optimisation**.
- Improved fabrication efficiency by integrating CNC/robotic constraints into the design pipeline.
- **Prototyped real outputs using robotic arms and laser cutters**.
- Presented at IAAC digital fabrication and sustainability reviews.
- Designed a **modular timber structure** using **parametric modelling in Grasshopper** to reduce offcut waste by **22%** in fabrication simulations.
- Conducted **life cycle carbon assessment** showing a **45% lower embodied CO₂** compared to standard CLT assemblies.
- Coordinated with the fabrication lab to generate **CNC-ready cut files** from parametric definitions, decreasing prep time from 6 hours to **under 2 hours**.

AUGUST 2022-FEBRUARY 2023

Architectural Intern | K12 Atelier | Delhi

- Worked on a 4-storey commercial complex from design to site coordination. Supported **redlining, 3D modelling, and presentation decks for client meetings**.
- Assisted in a competition submission shortlisted in the top 10 of 100+ entries.

2023 - 2025

Master in Advanced Architecture (MAA):

EDUCATION

Institute for Advanced Architecture of Catalonia Barcelona, Spain

- Activities and societies: – Studio Research Cluster: Computational Design. Public Review Panel Presenter & Material Systems Lab (Bioplastics + Robotics).
- Completed my **Master's in Advanced Architecture at IAAC** with a **specialisation in Algorithmic and Computational Design**. My academic focus centred on the intersection of urban mobility, data-driven design, and AI in architecture.

2018 - 2023

Bachelor of Architecture (B.Arch):

Mysore School of Architecture Mysore, India

- Activities and societies: Participated in multiple design competitions and studio projects exploring **urban informality, heritage preservation, and typology adaptation**.
- Completed a **5-year B.Arch** with a strong foundation in **spatial design, construction systems, and architectural theory**.
- The final-year thesis explored **Football stadium Design** for the Kerala Blasters team in Kochi.

LANGUAGES

English



Hindi



Spanish



ACHIEVEMENTS

Best Project Award – AI in Architecture Studio IAAC (2024) – Recognized for the WOOD-ID initiative, improving wood efficiency by 25%.

PUBLICATION

Communique – International Conference on Urban Villages: A Transformational Discourse in Architecture and Planning (22–23 May 2022)

Published by Urban Village Charitable Trust (UVCT), New Delhi.